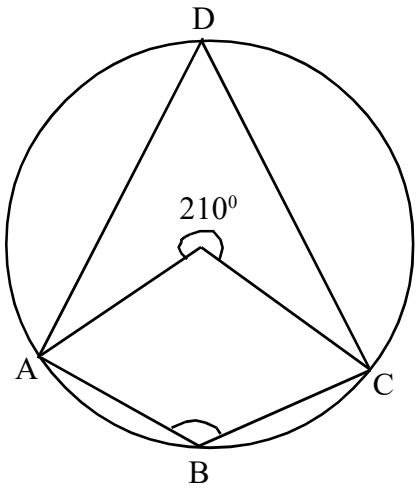
Chapter 2

CIRCLES

1. In the figure central angle of an arc is 210^0. $\angle B$ and $\angle D$ are the angles on the alternate arc.



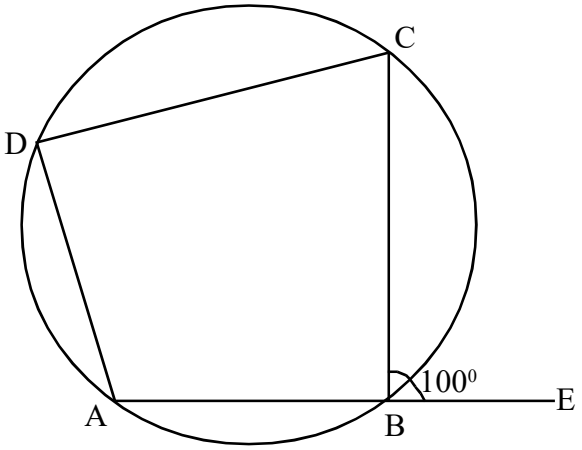
What is the measure of $\angle B$?

A) 75^0 B) 105^0

C) 210^0 D) 110^0

2. In the figure, $\angle CBE = 100^0$.

Which among the following can be $\angle A + \angle D$?



A) 200^0 B) 80^0

C) 100^0 D) 290^0

3. Which among the following is not a cyclic quadrilateral.

A) Rectangle B) Square

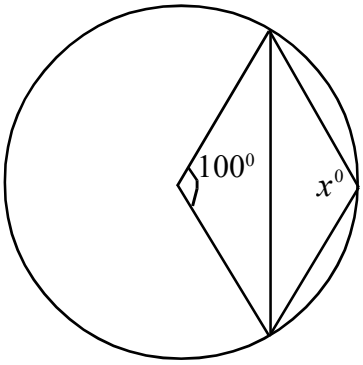
C) Isosceles Trapezium D) Parallelogram

4. In the figure O is the centre of the circle.

What is the value of x?

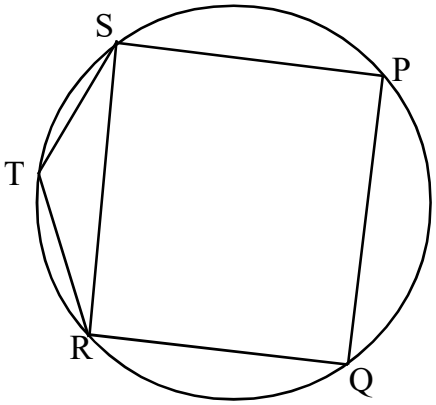
A) 130^0 B) 140^0

C) 105^0 D) 120^0



5. In the figure, PQRS is a square. What is the measure of $\angle STR$?

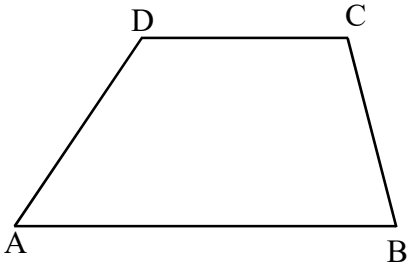
(120^0, 155^0, 135^0, 110^0)



6. Read the two statements given below.

Statement 1 : In quadrilateral ABCD the sides AB and DC are parallel and $\angle A \neq \angle B$. ABCD is a cyclic quadrilateral.

Statement 2 : Non-isosceles trapezium are not cyclic.



A) Statement 1 is true and statement 2 is false

B) Statement 1 is false and statement 2 is true

C) Both Statements are true and statement 2 is the reason of statement 1.

D) Both Statements are true and statement 2 is not the reason of statement 1

7. Read the two statements given below.

Statement 1 : Sum of outer angles of opposite vertices of a cyclic quadrilateral is 180^0.

Statement 2 : In a cyclic quadrilateral, outer angle at a vertex is equal to inner angle at the opposite vertex.

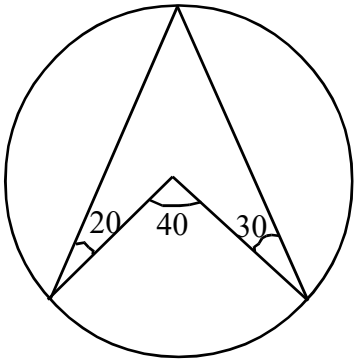
A) Statement 1 is true and statement 2 is false

B) Statement 1 is false and statement 2 is false

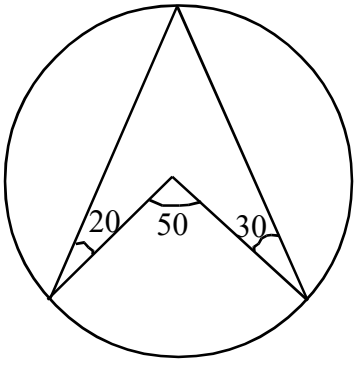
C) Both Statements are true and statement 2 is the reason of statement 1.

D) Both Statements are true and statement 2 is not the reason of statement 1

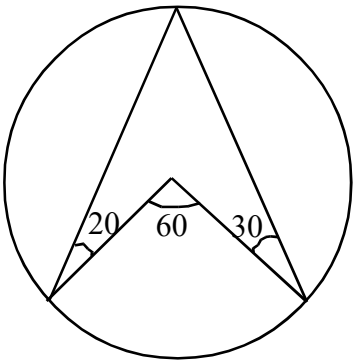
8. Which among the following figure, the angles are marked correctly?



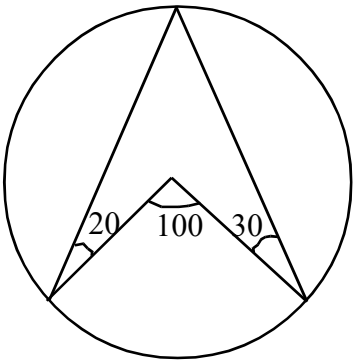
A



B



C

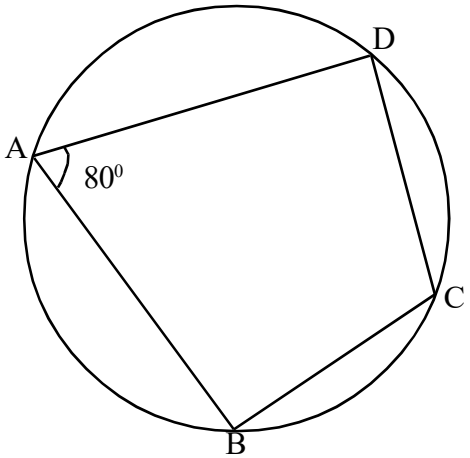


D

9. In the figure, $\angle A = 80^0$.

What is the measure of central angle of arc BAD?

(100^0, 200^0, 80^0, 160^0)



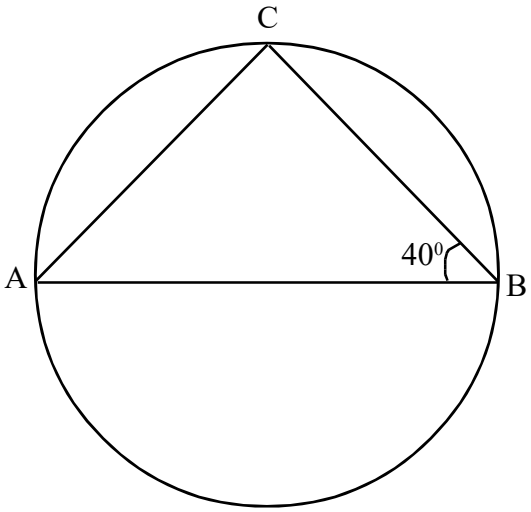
10. What is the measure of central angle of $\frac{1}{4}$ part of a circle.

(360^0, 120^0, 90^0, 60^0)

11. Read the statements below:

Statement 1 : In the figure, AB is the diameter. If $\angle B = 40^0$ then, $\angle A = 50^0$.

Statement 2 : The angle in a semicircle is a right angle.



A) Statement 1 is true and statement 2 is false

B) Statement 1 is false and statement 2 is false

C) Both Statements are true and statement 2 is the reason of statement 1.

D) Both Statements are true and statement 2 is not the reason of statement 1

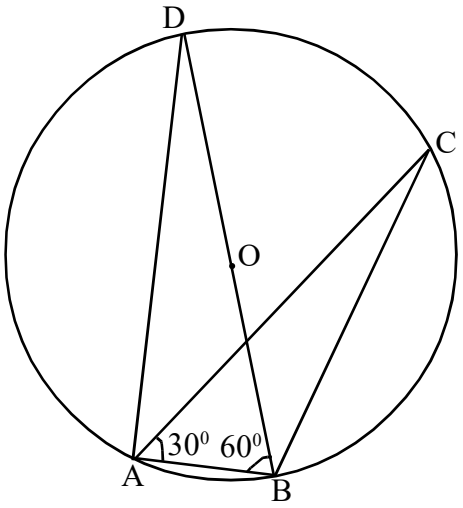
12. In the figure O is the centre of the circle.

A, B, C, D are points on the circle.

$\angle CAB = 30^0$, $\angle ABD = 60^0$

i) $\angle ADB$ = ...

ii) $\angle ABC$ = ...



13. Angles on the alternate segments of a circle are in the ratio 1:3.

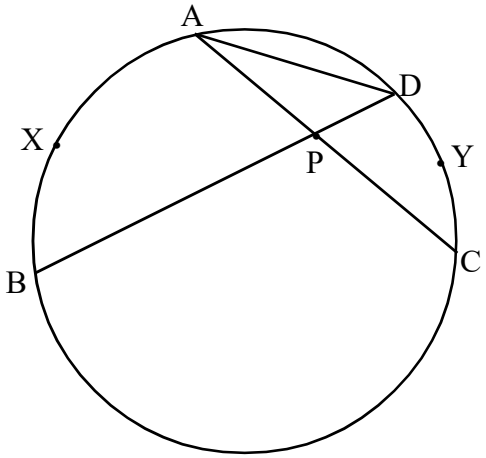
What is the measure of angles in each segment?

14. In the figure, central angle of arc AXB is 60^0 and arc CYD is 70^0.

i) Find the measure of $\angle D$?

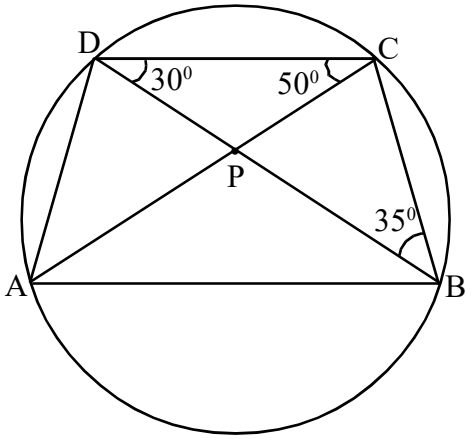
ii) Find the measure of $\angle A$?

iii) Find the measure of $\angle APB$?



15. i) Find the angles of cyclic quadrilateral ABCD in the figure.

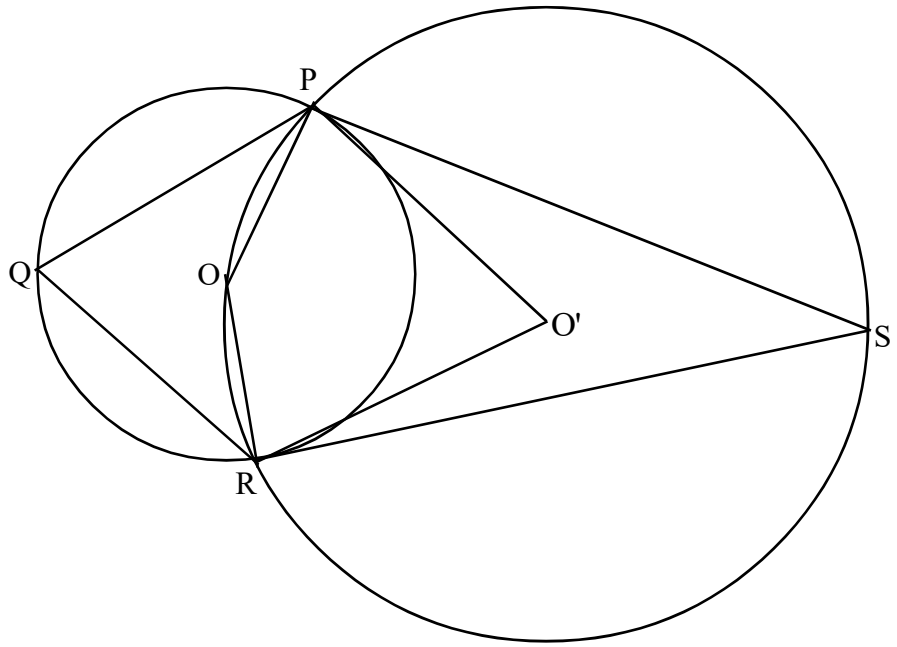
ii) Find all angles of ΔBPC ?



16. Draw a circle of any radius. Draw a chord such that angle on one segment should be three and half times the angle on the other segment.

17. Draw a triangle with circumradius 3.5 cm and two of the angles 35^0 and 72^0.

18.



In the figure, O and O' are the centre of the two circles which are intersecting at the points P and R. $\angle PQR = 80^0$.

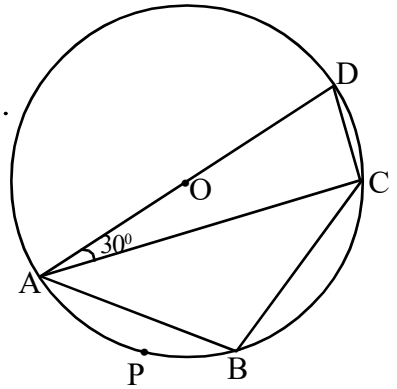
i) $\angle POR = ...$

ii) $\angle PSR = ...$

iii) $\angle PO'R = ...$

19. In the figure AD is the diameter of the circle.

The central angle of arc APB is 80^0 and $\angle DAC = 30^0$.

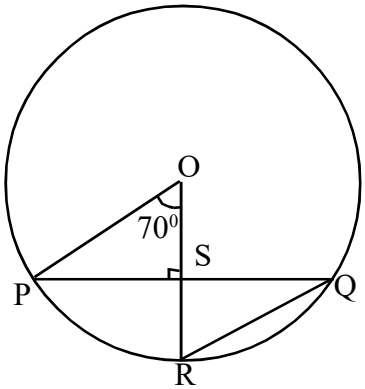


i) What is the measure of $\angle ACB$?

ii) What is the measure of $\angle ABC$?

iii) What is the measure of $\angle CAB$?

20.

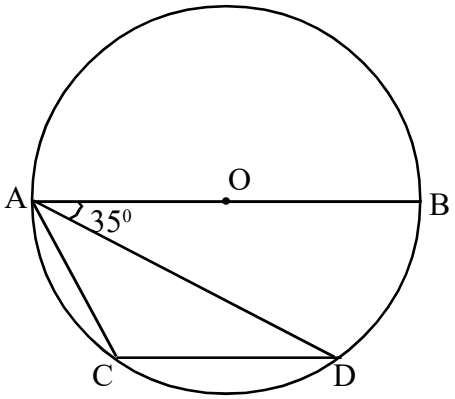


In the figure P, Q, R are the points on the circle with centre O. $\angle POR = 70^0$ and $\angle PSR = 90^0$

i) Find all the angles of ΔPOS .

ii) Find the all the angles of ΔRSQ .

21. In the figure chord CD is parallel to the diameter AB and $\angle BAD = 35^0$. Find all the angles of ΔACD.



22. In the figure A, B, C, D, E are the points on the circle with centre O and OC is parallel to DE. If $\angle OAB = 50^0$, $\angle OCB = 60^0$ and $\angle EDC = 35^0$.

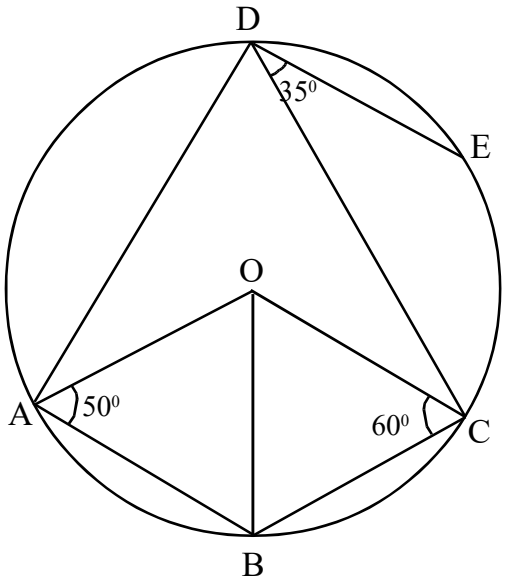
Find the measure of,

i) $\angle ABO$ = ...

ii) $\angle ADC$ = ...

iii) $\angle DAB$ = ...

iv) $\angle DAO$ = ...



23. ABCD is a cyclic quadrilateral.

$\angle A + \angle D = 210^0$ and $\angle D + \angle C = 250^0$.

i) Find $\angle A + \angle C$?

ii) Find the measure of $\angle A$ *and* $\angle C$?

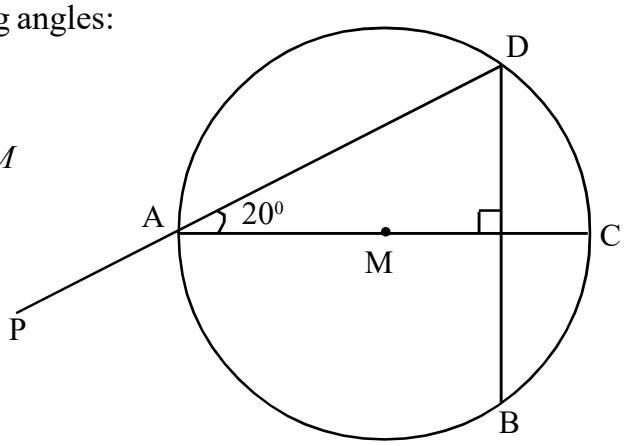
24. In the figure, Chord BD is perpendicular to diameter AC.

Find the measure of following angles:

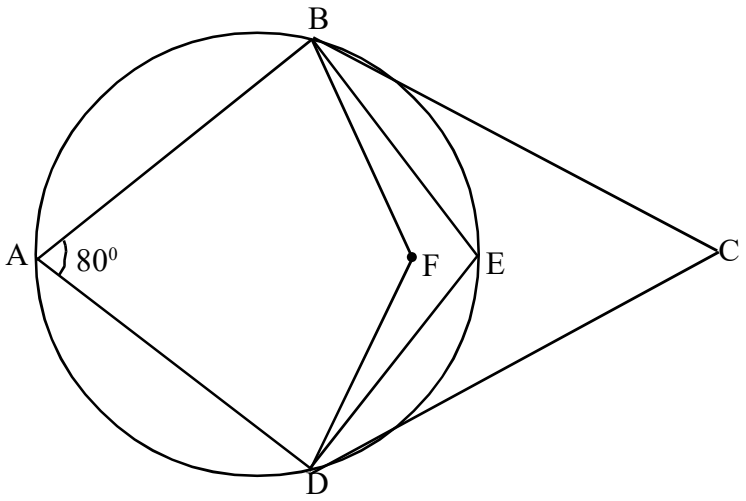
i) $\angle BAC$ ii) $\angle BCD$

iii) $\angle ADC$ iii) $\angle CDM$

v) $\angle BAP$



25. In the figure A, B, E, D are the points on the circle and $\angle A = 80^0$.



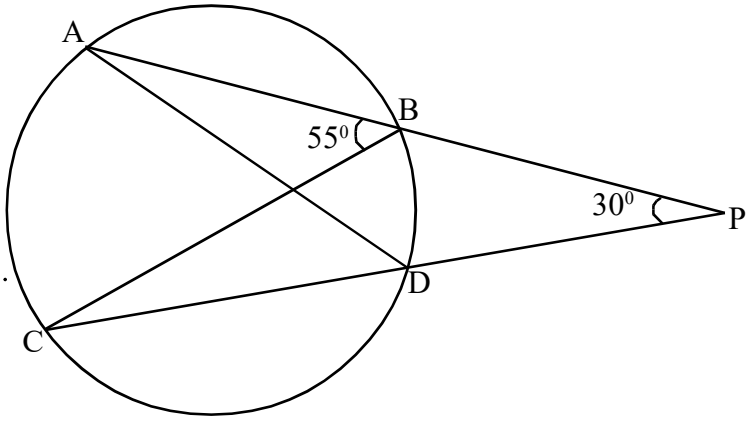
i) $\angle BED$ = ...

ii) $\angle BED + \angle BFD + \angle BCD = 270^0$

and the measure of smaller angle among these is 50^0. What is the measure of $\angle BFD$ *and* $\angle BCD$?

26. In the figure chords AB and CD extended to meet at the point P.

$\angle ABC = 55^0$ and $\angle APC = 30^0$.



i) $\angle BCD$ = ...

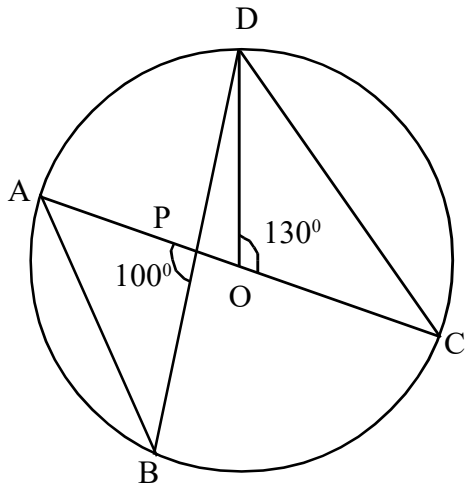
ii) $\angle BAD$ = ...

27. In the figure A, B, C, D are the points on the circle with centre O. Find the measure of following angles.

i) $\angle DBC$

ii) $\angle ACB$

iii) $\angle ABD$

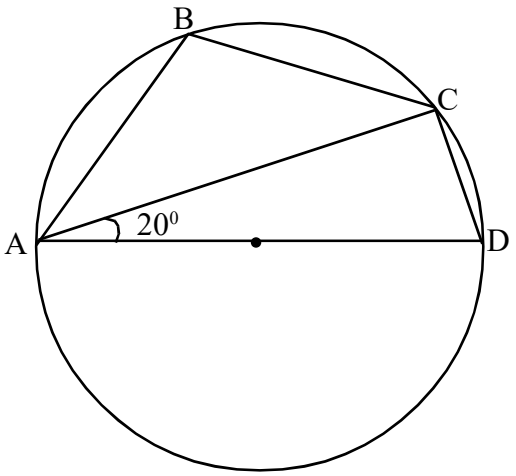


28. In the figure AD is the diameter of the circle and length of the chords AB and BC are equal.

$\angle CAD = 20^0$.

i) Find the measure of $\angle ADC$?

ii) Find the measure of $\angle BAD$?

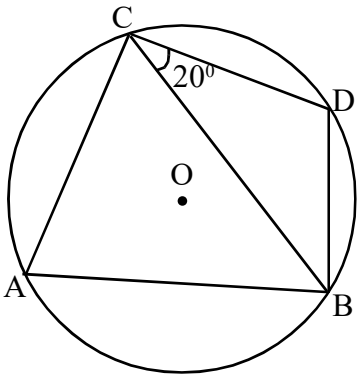


29. In the figure, ABCD is a cyclic quadrilateral and length of chord CD is equal to the radius of the circle. $\angle BCD = 20^0$

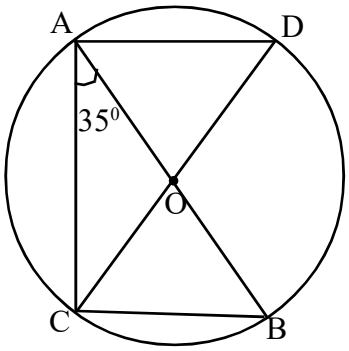
i) What is the measure of central angle of chord CD?

ii) What is the measure of $\angle CBD$?

iii) What is the measure of $\angle CAB$?



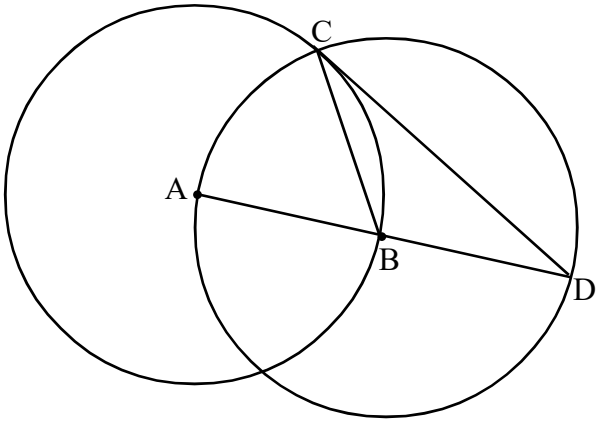
30. In the figure A, B, C, D are the points on the circle with centre O and $\angle BAC = 35^0$.



i) $\angle ACB$ = ...

ii) $\angle ADC$ = ...

31.



In the figure, A and B are the centre of two circles of same radius.

i) What is the measure of $\angle ABC$?

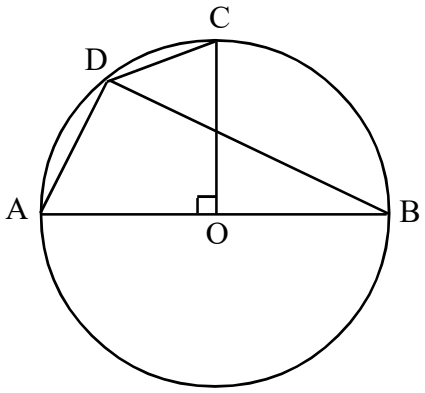
ii) What is the measure of $\angle BDC$?

32. In the figure, radius OC is perpendicular to the diameter AB.

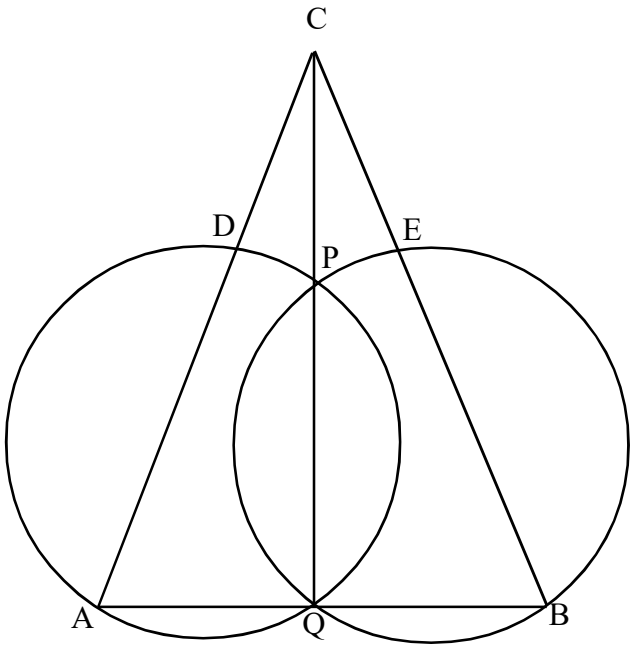
D is a point on the circle.

i) Find $\angle BDC$?

ii) Find $\angle ADC$?



33.



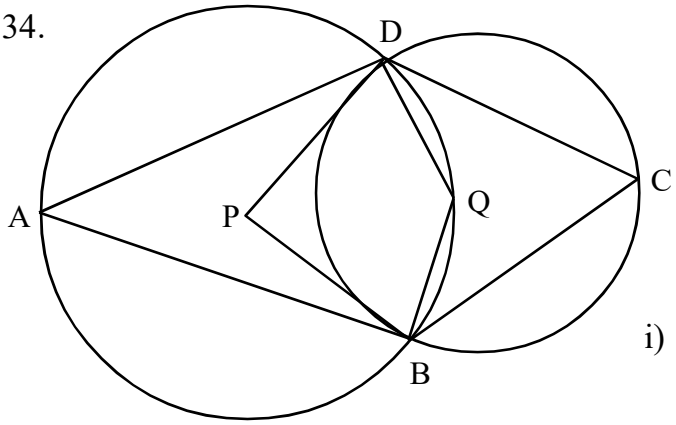
In the figure, two circles intersect at the P and Q. The chords AD and BE are extended to meet at the point C.

i) If $\angle AQP = x^0$, what is $\angle ADP$?

ii) Prove that $\angle ADP + \angle BEP = 180^0$.

iii) Prove that CDPE is a cyclic quadrilateral.

34.

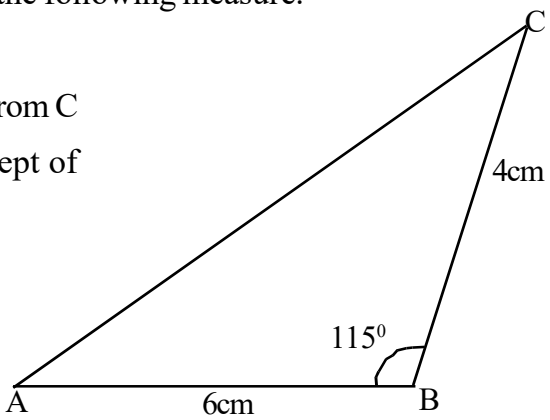


In the figure, P and Q are the centre of the two circles and $\angle BAD = 30^0$. Find the measure of the following angles.

i) $\angle BPD$ ii) $\angle BQD$ iii) $\angle BCD$

35. Draw triangle ABC with the following measure.

Draw the perpendicular from C to AB by using the concept of angle on a semi circle.

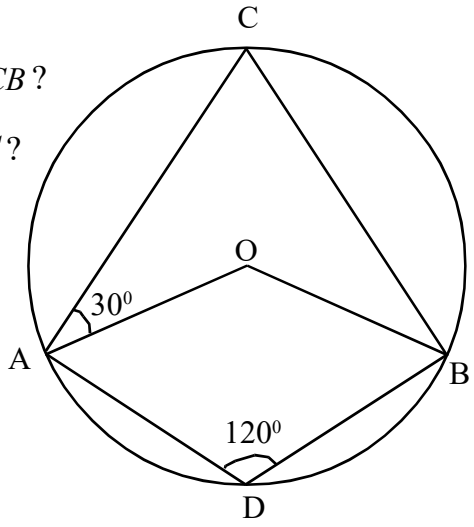


36. In the figure, O is the centre of the circle.

$\angle ADB = 120^0$, $\angle OAC = 30^0$

i) What is the measure of $\angle ACB$?

ii) Find all the angles of ΔABC ?

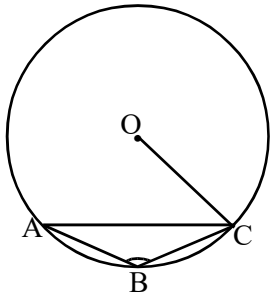


37. A, B, C are the points on a circle with centre O.

If $\angle OCA = x^0$,

i) What is $\angle OAC$?

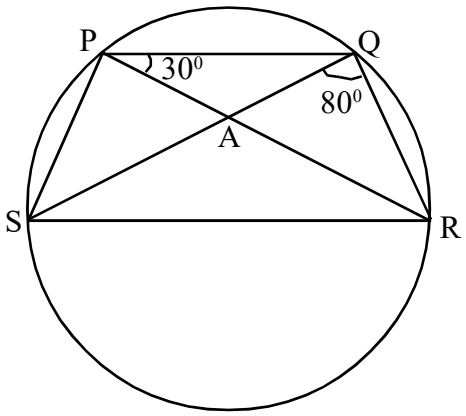
ii) Prove that $\angle ABC - \angle OCA = 90^0$?



38. Draw a triangle with circum radius 3cm and two of the angles $27½^0$ and $32½^0$?

39. In the figure, PQRS is a cyclic quadrilateral in which diagonals PR and QS intersects at A. $\angle SQR = 80^0$ and $\angle QPR = 30^0$

Find the measure of $\angle SRQ$?



Chapter 2

CIRCLES

ANSWERS

1. B
2. A
3. D
4. A
5. 135^0
6. B
7. C
8. D
9. 200^0
10. 90^0
11. C
12. i) $\angle ADB = 30^0$

    ii) $\angle ABC = 120^0$
13. Sum of angles in alternative segment is 180^0

    Angle in a segment $= \frac{1}{4} \times 180^0 = 45^0$

    Angles in alternative angle $= \frac{3}{4} \times 180^0 = 135^0$
14. i) $\angle D = \frac{60^0}{2} = 30^0$

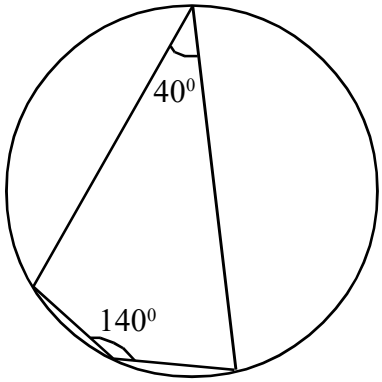
    ii) $\angle A = \frac{70^0}{2} = 35^0$

    iii) $\angle APB = \angle A + \angle D = 35^0 + 30^0 = 65^0$

15. i) $65^0,\ 85^0,\ 115^0,\ 95^0$

ii) $35^0,\ 65^0,\ 80^0$

16.



17. Construction of triangle

18. i) $\angle POR = 160^0$ ii) $\angle PSR = 20^0$ iii) $\angle PO^1R = 40^0$

19. i) $\angle ACB = 40^0$ ii) $\angle ABC = 120^0$ iii) $\angle CAB = 20^0$

20. Angles of ΔPOS are $20^0, 70^0, 90^0$

Angles of ΔRSQ are $35^0, 55^0, 90^0$

21. $\angle ADC = 35^0$, BD Joined $\angle ABD = 55^0$

$\therefore \angle ACD = 125^0$

$\angle DAC = 180^0 - (35^0 + 125^0) = 180^0 - 160^0 = 20^0$

22. i) $\angle ABO = 50^0$

ii) $\angle ADC = 70^0$

iii) $\angle DAB = 85^0$

iv) $\angle DAO = 35^0$

23. $\angle C - \angle A = 40^0$

i) $\angle C + \angle A = 180^0$

ii) $\angle C = \frac{180+40}{2} = \frac{220}{2} = 110^0$

$\angle A = \frac{180-40}{2} = \frac{140}{2} = 70^0$

24. i) $\angle BAC = 20^0$

ii) $\angle BCD = 140^0$

iii) $\angle ADC = 90^0$

iv) $\angle CDM = 70^0$

v) $\angle BAP = 140^0$

25. i) $\angle BED = 100^0$

ii) $\angle BFD = 120^0$, $\angle BCD = 50^0$

26. i) $\angle BCD = 25^0$

ii) $\angle BAD = 25^0$

27. i) $\angle DBC = \frac{1}{2}\angle DOC = \frac{130}{2} = 65^0$

ii) $\angle ACB = 100 - \angle DBC = 100^0 - 65^0 = 35^0$

iii) $\angle ABD = 25^0$

28. i) $\angle ADC = 70^0$

ii) $\angle BAD = 55^0$

29. i) $\angle COD = 60^0$

ii) $\angle CBD = \frac{1}{2}\angle COD = 30^0$

iii) $\angle CAB = 50^0$

30. i) $\angle ACB = 90^0$

ii) $\angle ADC = \angle ABC = 55^0$

31. i) $\angle ABC = 60^0$

ii) $\angle BDC = \frac{1}{2}\angle ABC = 30^0$

32. i) $\angle BDC = \frac{1}{2}\angle BOC = 45^0$

ii) $\angle ADC = \angle ADB + \angle BDC = 90^0 + 45^0 = 135^0$

33. i) $\angle ADP = 180 - x^0$

34. i) $\angle BPD = 60^0$ ii) $\angle BQD = 150^0$ iii) $\angle BCD = 75^0$

35. Construction of triangle

36. i) $\angle ACB = 60^0$

ii) 60^0 , 60^0 , 60^0

37. i) $\angle OAC = x^0$

38. Construction of triangle

39. $\angle SRQ = 70^0$